

G LLM Prompts

This section provides all prompts used throughout the experiments. Summarization ([Listings 1 and 2](#)), question generation ([Listing 3](#)), question answering ([Listing 4](#)), answer claim splitting ([Listing 5](#)), and introspection ([Listing 6](#)).

Listing 1: Summarization prompt

```
## Document
{{ text }}

## Instruction
Please summarize the above document. Use up to {{ length_target }} words. Respond exactly in following JSON format:

{
  "summary": "(the {{ length_target }} words summary)"
}
```

Listing 2: Summarization prompt for meeting transcripts

```
## Meeting Transcript
{{ text }}

## Instruction
Please summarize the above meeting transcript. Use up to {{ length_target }} words. Respond exactly in following JSON format:

{
  "summary": "(the {{ length_target }} words summary)"
}
```

Listing 3: Question generation prompt

```
Your task is to analyze summaries of different lengths within a given genre. Your goal is to create question-answer pairs that capture the essence of information typically included in various summary lengths. Below is the dataset where each document was summarized in 5 different lengths.

# Dataset

{% for document in documents %}
## Document {{ loop.index }}

{% for length, text in document.items() %}
### Summary {{ length }}
{{ text }}

{% endfor %}
{% endfor %}

# Genre
The genre of the documents:
{% if dataset == "rct" %}
  Randomized controlled trials (RCT) in the clinical domain.
{% elif dataset == "astro-ph" %}
  Discussion section in astrophysics papers.
{% elif dataset == "cs-cl" %}
  Related work section in NLP papers.
{% elif dataset == "qmsum" %}
  Meeting transcripts.
{% endif %}

# Task
Each text in the dataset has been summarized in 5 different lengths (in 10, 20, 50, 100, and 200 words). Your task is to analyze the summaries and identify the types of information typically included at each summary length. To do this, please proceed as follows:

1. Carefully read the summaries, paying attention to what information is included or omitted.
2. For each summary length, create a set of question-answer pairs that represent typical information included at this length. The questions should be general enough to apply to many documents in this genre, while the answers will naturally be different across documents.
```

Important guidelines:

- Ensure that your questions are relevant to the genre and capture information that would be commonly found in texts of this type.
- It is really important that the questions are answerable with most documents in this genre, not just with the ones presented here! To this end, state a prototypical answer to each question.
- The questions should be unique to each length. That means, do not repeat a question if it is already sufficiently covered at the shorter length.
- Start with question words (What, How, Why, Which) rather than 'Can you'
- Make the topic the grammatical subject of a question.
- Keep the questions concise and focused.
- Create at least 3-5 questions for each summary length.

Structure your response as a valid json object with the following format:

```
{
  "questions_10_words": [
    {
      "question": "",
      "example_answer": "",
    }
  ],
  [... truncated for brevity ...]
}
```

Listing 4: Question answering prompt

Answer the following question given the text. If the question cannot be answered with the text, reply "no answer".

```
## Text
{{ text }}
```

```
## Question
{{ question }}
```

First, carefully read and analyze both the text and the question. Then provide the answer. Please follow these guidelines:

- If the question cannot be answered, reply with "no answer"
- Use only information explicitly stated in or directly implied by the text
- Do not include any external knowledge or personal opinions
- Aim for concise answers that include all important points relevant to the question

Please use this format for your response:

Question: [restate the question exactly]

Answer: [the answer based on the text or "no answer"]

Listing 5: Claim splitting prompt

You split sentences into a list of facts that we explicitly know from the sentence. Make each fact as atomic as possible.

Sentence: Protein-rich nutrition is necessary for wound healing after surgery.

Output:

```
[
  "Protein-rich nutrition is necessary for wound healing.",
  "Wound healing occurs after surgery."
]
```

Sentence: In this study, the benefit of preoperative nutritional support was investigated for non-small cell lung cancer patients who underwent anatomic resection.

Output:

```
[
  "The study investigated the benefit of preoperative nutritional support.",
  "The study considers patients with non-small cell lung cancer.",
  "The study considers patients who underwent anatomic resection."
]
```

[... 12 few-shot examples truncated for brevity ...]

Here is a new sentence. Please split it into a list of facts that we explicitly know from the sentence. Make each fact as atomic as possible. Output the facts as Python list. Only output the list, nothing more.

Sentence: {sent}
Output:

Listing 6: Introspection prompt

```
## Task
{% if dataset == "rct" %}
    You are a research expert in randomized controlled trials (RCTs). Imagine you are asked to summarize a
    paper describing the results of an RCT for a typical reader in this field. The summary should provide
    enough context to stand alone, since the reader will only see your summary and no other parts of the paper.
{% elif dataset == "astro-ph" %}
    You are a research expert in astrophysics. Imagine you are asked to summarize the discussion section
    of an astrophysics paper for a typical reader in this field. The summary should provide enough context to
    stand alone, since the reader will only see your summary and no other parts of the paper.
{% elif dataset == "cs-cl" %}
    You are a research expert in natural language processing (NLP). Imagine you are asked to summarize the
    related work section of an NLP paper for a typical reader in this field. The summary should provide enough
    context to stand alone, since the reader will only see your summary and no other parts of the paper.
{% elif dataset == "qmsum" %}
    You are an expert in communications and meetings. Imagine you are asked to summarize a meeting
    transcript (e.g., research group meetings) for a typical reader of these texts. The summary should provide
    enough context to stand alone, since the reader will only see your summary and not the full meeting
    transcript.
{% endif %}

The summary length is constrained, requiring you to think about what content to prioritize. Ask yourself:
what are some key questions you want the summary to answer? Your task is to rate the relative importance
of a list of questions that the summary could answer.

## Questions
Here is the list of questions you should evaluate.

{% for question in questions %}
{{ loop.index }}. {{ question }}
{% endfor %}

## Rating
Please use the following scale, going from least important to most important.

1 - Least important; I would exclude this information from a summary.
2 - Low importance; I would include this information if there is room.
3 - Medium importance; I would probably include this information.
4 - High importance; I would definitely include this information.
5 - Most important; One of the first questions to be answered in the summary.

For each rating, please provide a brief (1-sentence) rationale explaining your decision or highlighting
any considerations or uncertainties you had.

Important considerations:
- Use the full scale (1-5) to express relative importance.
- Remember that space in the summary is limited, so not everything can be included, and you CANNOT rate
  all questions as 5.
- Make sure to rate all given questions.

Please respond as a valid JSON list with following format:

[
  {
    "id": "[the question number]",
    "question": "[repeat the exact question]",
    "rationale": "[your one-sentence rationale for the rating]",
    "rating": "[your numeric rating, 1-5]"
  }
]
```